Shoppinglist Generator with Health Advice

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Problem

Every human-being has to go grocery shopping which also requires a large proportion of the person's income. Money can be saved by creating a shopping list in the first place to avoid impulse decisions in the store. However, a cheap grocery shopping trip may result in lower quality food and can impair the person's health. The decision between healthy food and a cheap purchase can be difficult. Especially for people with lower income which implies low quality food most of the time.

Idea

We want to create an app in which the user can create certain dishes with their ingredients and generate a shopping list based on the user's selection. The ingredients required for a certain dish contain the price, but also certain information about the "healthiness" of the product (E.g. "NutriScore", kCal, Data from the "Nutritional information" table (Nährwerttabelle)). The health data is then analyzed to provide user feedback about the planned shopping trip as well as the expected price.

- **Client Only**: Recipe Storage, Selection, Shopping List and Calculations all happening locally for privacy.
- **Recipe Storage** holds all the users dishes with their ingredients which are either created manually or pulled from common public APIs for cooking recipes.
- **Health Evaluation** takes all the information from the single ingredients currently selected for the shopping list and analyzes them. The is done with deterministic algorithms and certain thresholds to decide what degree of healthiness the shopping list or a particular dish has. (Possible AI Integration)
- **Possible Server Integration**: Global database with nutritional information and price for ingredients the clients can pull from.
- **Recommendations**: The app makes dish recommendations for the user based on previous shopping list generations in order to be varied and healthy.

[Main components and their interaction. Architectural style or patterns and why. Driving quality attributes. Out of scope: . . .]

Benefits

- [Benefit 1 – addresses . . . ; demonstrated by . . .]
- [Benefit 2 – addresses . . . ; demonstrated by . . .]

Action

1. ...
2. ...

[Required technologies, data, and infrastructure; team split.]

AI usage statement. None [Which AI tools and models did you use, and for what (e.g. brainstorming, rephrasing)? Write "None" if you did not use AI.]